



Utilities & Caching (`app/utls/`)

Coresight Research Portal — Technical Documentation

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Application: Market Data Portal (MDP)

Scope: Logging, Streamlit cache management, Azure Blob, Portable Document Format (PDF) generation, email dispatch, market data helpers, background scanning, local storage, constants

Overview

`app/utils/` provides cross-cutting infrastructure used by pages, core modules, data services, and components. Every module imports `server_logger` instead of `stdlib logging`. Each module exposes a consistent Application Programming Interface (API).

Figure 1 Part A — Startup triggers and core utils

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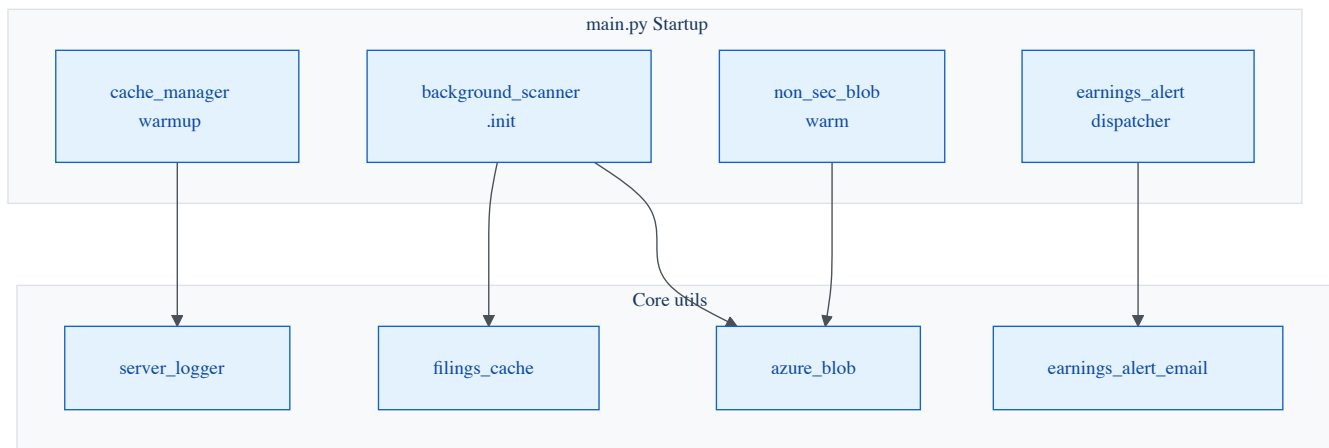
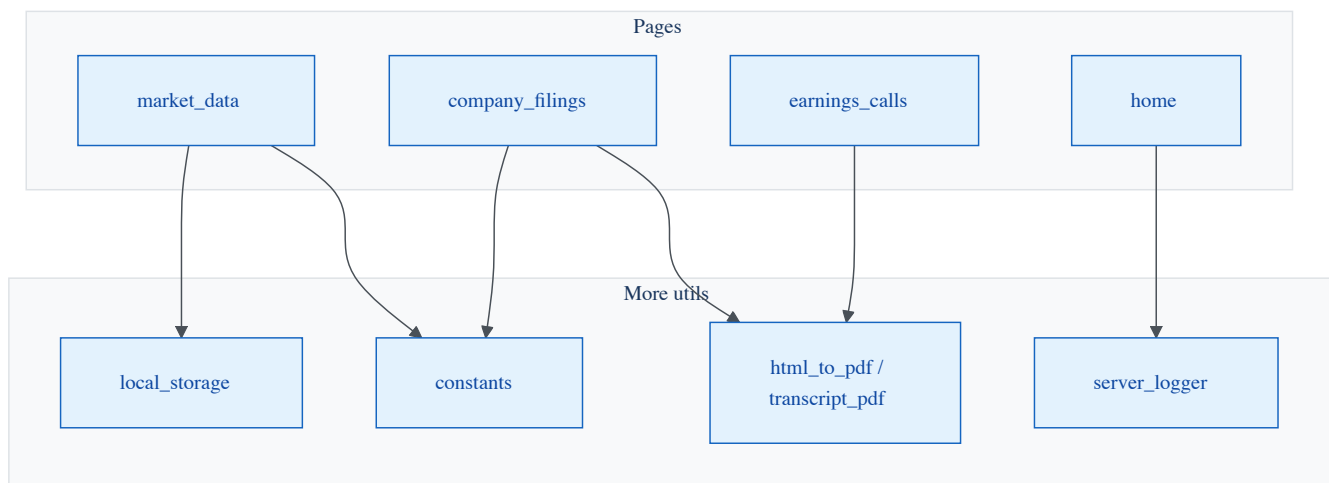


Figure 1 Part B — Pages consume utils

Figure 1 Part B — Pages consume utils



Module Index

Module	Status	Purpose
<code>server_logger.py</code>	Core	Central application logging
<code>cache_manager.py</code>	Active	Streamlit cache control + background warmup
<code>azure_blob.py</code>	Active	Azure Blob Storage helpers
<code>filings_cache.py</code>	Active	Persistent blob metadata cache
<code>background_scanner.py</code>	Active	Background Azure filing pre-fetch
<code>non_sec_blob_fallback.py</code>	Active	Non-U.S. Securities and Exchange Commission (SEC) filing local index
<code>html_to_pdf.py</code>	Active	SEC filing PDF (PyMuPDF)
<code>transcript_pdf.py</code>	Active	Earnings transcript PDF (fpdf2)
<code>earnings_alert_email.py</code>	Active	Simple Mail Transfer Protocol (SMTP) earnings alert emails
<code>earnings_alert_dispatcher.py</code>	Active	Background alert dispatch
<code>excel_export.py</code>	Active	Branded Excel export
<code>local_storage.py</code>	Active	Session-state preference keys
<code>local_storage_manager.py</code>	Active	Browser persistence via streamlit-local-storage
<code>ticker_utils.py</code>	Active	Ticker validation + fallback
<code>constants.py</code>	Active	Currency, exchange, country, topic lookups
<code>filing_units.py</code>	Active	Multi-filing document support (8-K, 6-K, etc.)
<code>retailer_forecaster.py</code>	Active	Revenue forecasting engine
<code>cache_cleaner.py</code>	Active	<code>__pycache__</code> purge on rerun
<code>blob_doc_type_discovery.py</code>	Standalone tool	Azure doc_type gap scanner
<code>filing_pdf.py</code>	Dead	Replaced by <code>html_to_pdf.py</code>
<code>market_data.py</code>	Deprecated	Legacy duplicate of <code>pages/market_data.py</code>
<code>companyProfile.py</code>	Dead	Replaced by <code>components/companyProfile.py</code>
<code>__init__.py</code>	Package marker	—

server_logger.py — Central Logging

Output: server-logs/server-log.log

Format: {timestamp} | {level} | {rerun_id} | page={page} | {filename}:{funcName} | {message}

Core API

Category	Functions
Rerun correlation	new_rerun_id , get_rerun_id , set_page_context
Log levels	log_info , log_debug , log_warning , log_error , log_exception , log_critical , log_structured_error
Error handling	error_boundary , safe_execute
Performance	log_timing , log_db_timing , log_render_timing , log_function_timing
Context managers	timed_operation , timed_db_query , timed_render , timed_function
Page tracking	PageTimer , PageLoadTracker , NavigationTracker , track_page_load
Log admin	get_log_content , clear_logs , download_logs , get_log_stats , analyze_slow_operations

Environment variables

Variable	Default	Role
APP_LOG_LEVEL	WARNING	Minimum log level
APP_TIMING	1	Timing-only filter mode
APP_VERBOSE_LOGGING	0	Verbose output

Rule: Stdlib logging is forbidden project-wide. All code imports from utils.server_logger only.

cache_manager.py — Streamlit Cache Control

Purpose

Centralized `@st.cache_data` management: clear, stats, foreground/background warming for news, companies, earnings, Yahoo Finance (YF) topics.

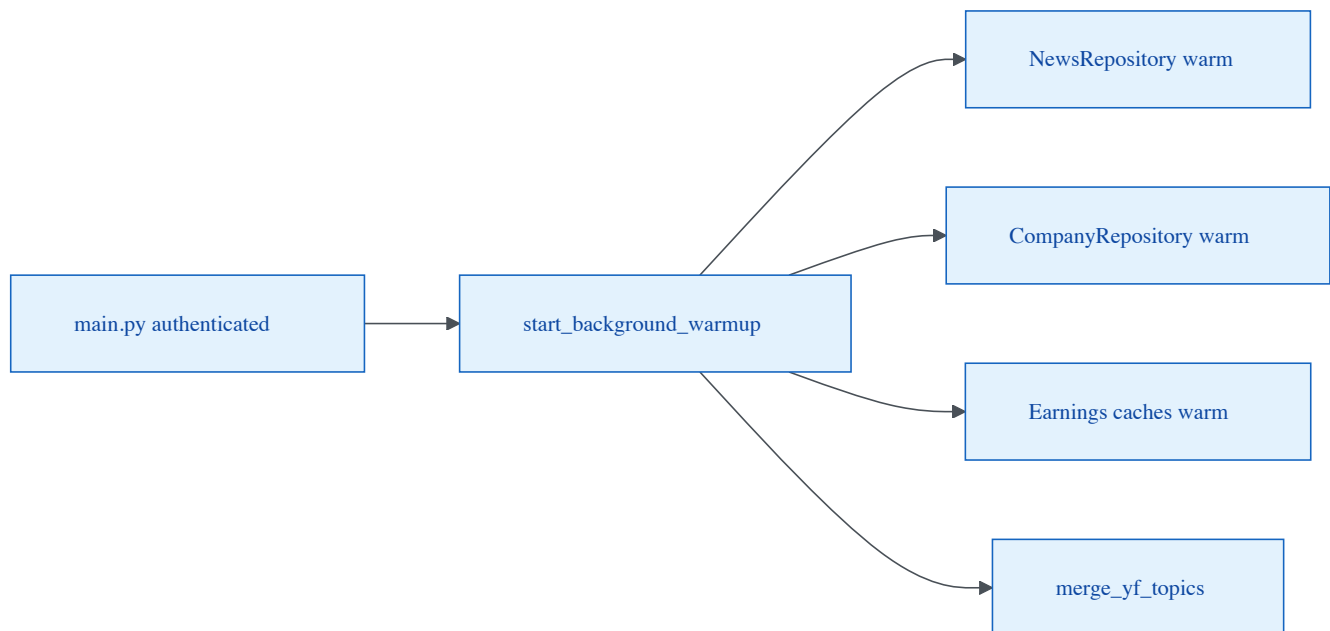
API

Function	Role
<code>clear_news_cache()</code> / <code>clear_company_cache()</code> / <code>clear_all_caches()</code>	Cache invalidation
<code>get_cache_stats()</code>	Hit/miss statistics
<code>warm_caches()</code>	Foreground warmup
<code>start_background_warmup()</code>	Daemon thread warmup
<code>render_cache_controls()</code>	Optional admin User Interface (UI)

Startup integration

Called from `main.py` when authenticated and `ENABLE_BG_WARMUP=1` (default).

Called from main.py when authenticated and ENABLEBGWARMUP=1 (default).



Warms: `data.repository.NewsRepository`, `CompanyRepository`, earnings caches; calls `utls.constants.merge_yf_topics`.

Azure Blob & Filing Cache Subsystem

Architecture

Cross-cutting utilities consumed by pages, core modules, and background warmups.

Figure 2 Part A — Azure source & utils modules

Figure 2 Part A — Azure source & utils modules

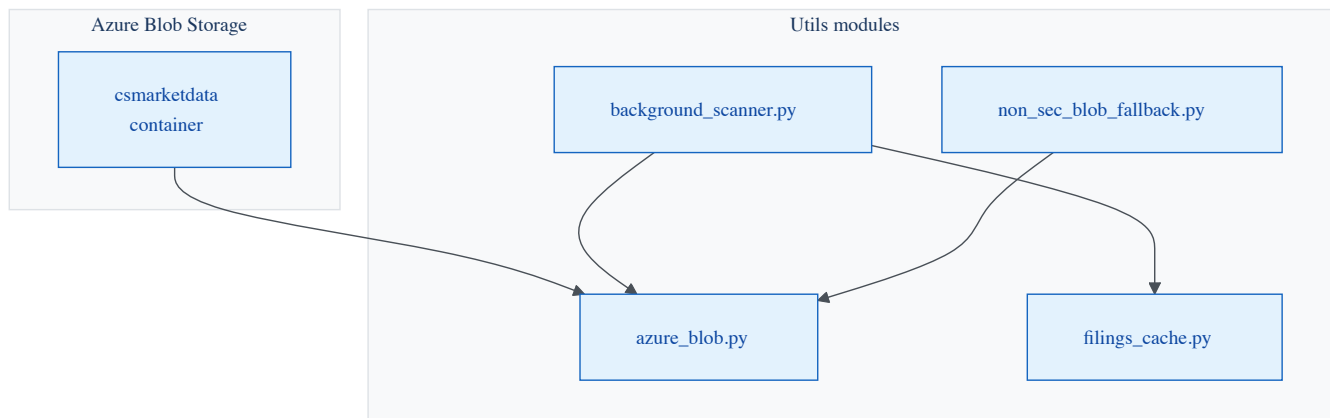
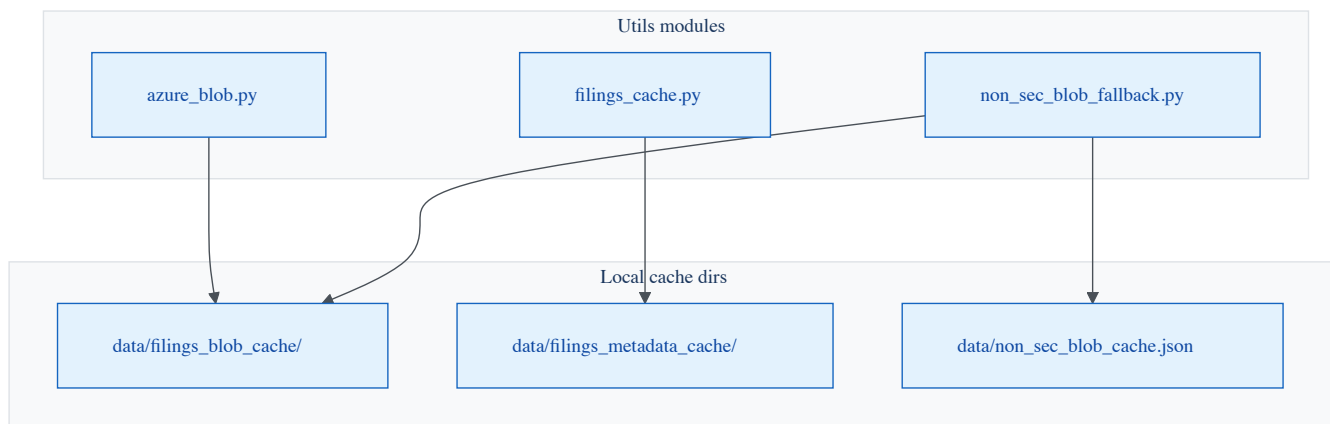


Figure 2 Part B — Local cache paths

Figure 2 Part B — Local cache paths



azure_blob.py

Function	Role
<code>get_container_client()</code>	Azure SDK container client
<code>local_cache_path_for_blob(blob_name)</code>	Local filesystem path
<code>ensure_local_blob_for_transcript(blob_name)</code>	Download if missing
<code>download_blob_to_memory(blob_name)</code>	In-memory bytes

Env vars:

Variable	Default
<code>AZURE_STORAGE_ACCOUNT_NAME</code>	<code>csmarketdata</code>
<code>AZURE_STORAGE_ACCOUNT_KEY</code>	(required)
<code>AZURE_BLOB_CONTAINER</code>	<code>azure-storage-test</code>
<code>AZURE_BLOB_PREFIX</code>	(optional prefix)

Cache root: `{repo}/data/filings_blob_cache/`

Consumers: `earnings_calls.py` , `non_sec_blob_fallback.py`

filings_cache.py

Class / Function	Role
<code>FilingsMetadataCache</code>	DiskCache/SQLite-backed metadata (24h TTL, ETag validation)
<code>InMemoryFallbackCache</code>	Fallback when DiskCache unavailable
<code>get_filings_cache()</code>	Singleton accessor
<code>generate_etag_from_blobs(blobs)</code>	ETag for cache invalidation
<code>invalidate_company_cache(ticker)</code>	Per-ticker purge

Consumers: `background_scanner.py` , `company_filings.py`

background_scanner.py

Purpose: Intelligent background pre-fetch of Azure filing blobs. Skips scan when cache sufficient; discovers years dynamically; parallel downloads.

Function	Role
<code>BackgroundFilingsScanner</code>	Main scanner class
<code>init_background_scanner(auto_start=True)</code>	Start daemon scanner
<code>ensure_cache_purge()</code>	One-time cache purge (<code>.cache_purge_v1_done</code>)
<code>get_scan_progress()</code> / <code>get_cache_audit()</code>	Progress for UI
<code>is_scan_complete()</code>	Completion flag

Startup: `main.py` when `ENABLE_BG_SCANNER=1` (default) and authenticated.

Consumers: `main.py` , `home.py` (progress UI), `company_filings.py`

Data dirs: `data/filings_blob_cache/` , `data/filings_metadata_cache/` , `.blob_container_last_modified`

`non_sec_blob_fallback.py`

Purpose: Local-first index of non-SEC filing PDFs (annual reports, half-yearly, interim). Azure only as fallback.

Function	Role
<code>get_non_sec_blob_data(ticker)</code>	Blob listing for ticker
<code>warm_non_sec_cache_at_startup(tickers, max_workers=32)</code>	Parallel warm
<code>get_non_sec_tickers_from_db()</code>	Tickers with non-SEC sources
<code>invalidate_non_sec_cache()</code>	Called when scanner downloads new blobs
<code>is_non_sec_source_company(ticker)</code>	Source check

Startup: `main.py` daemon thread when authenticated.

Consumer: `company_filings.py`

PDF Generation

html_to_pdf.py — SEC Filings (Production)

Function	Role
<code>generate_filing_pdf(html_content, title, ticker, ...)</code>	PyMuPDF (fitz) render to PDF bytes
<code>_clean_sec_html(html)</code>	Strip iXBRL/eXtensible Business Reporting Language (XBRL) junk, wrap in branded CSS

Consumer: `company_filings.py` (PDF download button)

transcript_pdf.py — Earnings Transcripts

Function	Role
<code>generate_transcript_pdf(call, segments, ...)</code>	fpdf2 A4 branded PDF

Consumer: `earnings_calls.py`

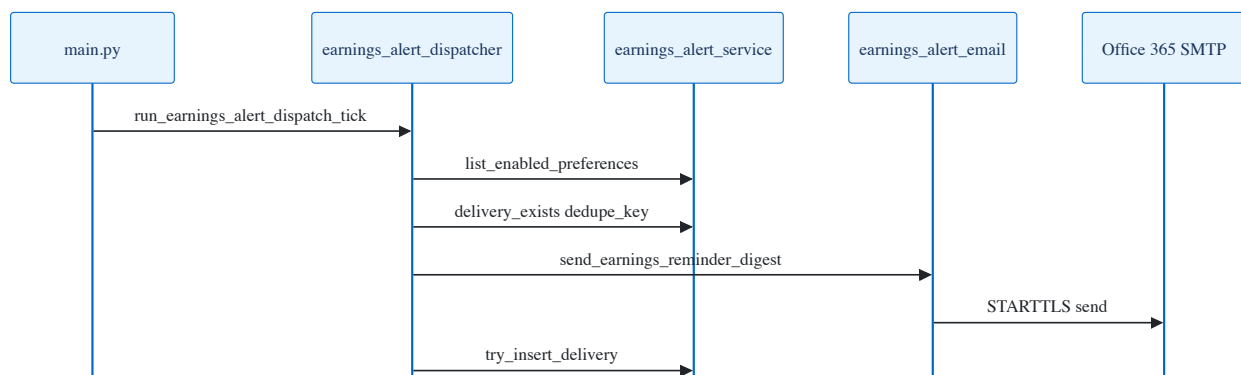
filing_pdf.py — Legacy (Dead)

Simple fpdf2 plain-text generator. **Superseded by** `html_to_pdf.py`. No consumers.

Earnings Alert Email Pipeline

Background dispatcher loads enabled preferences, dedupes deliveries, and sends SMTP alerts for upcoming earnings.

Background dispatcher loads enabled preferences, dedupes deliveries, and sends SMTP alerts for upcoming earnings.



earnings_alert_email.py

Function	Role
<code>send_earnings_alert_preferences_confirmation(...)</code>	User preference saved confirmation
<code>send_earnings_reminder_digest(...)</code>	Reminder email with upcoming earnings

Env vars:

Variable	Default
<code>SMTP_SERVER</code>	<code>smtp.office365.com</code>
<code>SMTP_PORT</code>	<code>587</code>
<code>FROM_EMAIL</code>	<code>dataautomation@coresight.com</code>
<code>EMAIL_PASSWORD</code>	Required — no-op if unset

earnings_alert_dispatcher.py

Function	Role
<code>run_earnings_alert_dispatch_tick()</code>	Throttled dispatch on each authenticated load
<code>start_background_dispatch_thread()</code>	Daemon thread for scheduled sends

Env vars:

Variable	Default
<code>EARNINGS_ALERT_DISPATCH</code>	<code>1</code>
<code>EARNINGS_ALERT_DISPATCH_THREAD</code>	<code>1</code>
<code>EARNINGS_ALERT_DISPATCH_MIN_INTERVAL_SEC</code>	<code>120</code>
<code>EARNINGS_ALERT_DISPATCH_HOUR_UTC</code>	<code>8</code>
<code>EARNINGS_ALERT_DISPATCH_LOOKBACK_DAYS</code>	<code>3</code>
<code>EARNINGS_ALERT_DISPATCH_THREAD_SLEEP_SEC</code>	<code>600</code>

Local Storage & State Persistence

`local_storage.py`

Session-state-backed typed keys via `StorageKey` enum.

Function	Stored value
<code>get/set_sec_filing(...)</code>	SEC filing filter state
<code>get/set_date_range(...)</code>	Date range picker
<code>get/set_marketdata_tab(...)</code>	Active Market Data tab
<code>get/set_selected_company(...)</code>	Global company selection
<code>clear_selected_company()</code>	Reset selection

Consumers: `market_data.py`, `home.py`, `components/toolbar.py`

`local_storage_manager.py`

Browser persistence via `streamlit-local-storage` under key `sip_app_state`.

Function	Role
<code>sync_market_data_state()</code> / <code>save_market_data_state()</code>	Market Data state round-trip
<code>get_persistent_state()</code> / <code>set_persistent_state()</code>	Generic namespaced state
<code>load_earnings_calls_state()</code> / <code>save_earnings_calls_state()</code>	Earnings Calls filters

Consumers: `market_data.py`, `home.py`, `earnings_calls.py`

Note: Degrades gracefully if `streamlit-local-storage` package missing.

`ticker_utils.py` — Ticker Validation

Function	Role
<code>validate_and_get_ticker(ticker, source)</code>	Database (DB) lookup; fallback to <code>AMZN</code> if invalid
<code>get_ticker_from_source(source)</code>	Extract ticker from URL/session
<code>check_ticker_exists(ticker)</code>	Boolean existence check

Constant: `DEFAULT_FALLBACK_TICKER = "AMZN"`

Consumers: `market_data.py`, `newsroom.py`, `company_filings.py`, `earnings_calls.py`, `earnings_calendar.py`

`constants.py` — Lookup Tables & Formatters

Key data exports

Constant	Contents
<code>CURRENCY_NAMES</code> / <code>CURRENCY_SYMBOLS</code>	ISO currency display
<code>EXCHANGE_CODE_MAP</code>	Exchange acronym → display name
<code>COUNTRY_NAMES</code>	Country code → name
<code>AV_TOPIC_LABELS</code> / <code>NEWS_TOPIC_LABELS</code>	News topic display labels
<code>SEGMENT_METRIC_GROUPS</code>	XBRL segment groupings for screening

Key functions

Function	Role
<code>get_currency_symbol(code)</code>	<code>\$</code> , <code>€</code> , etc.
<code>format_currency_full(value, code)</code>	Full formatted amount
<code>get_exchange_code(name)</code> / <code>format_exchange(code)</code>	Exchange formatting
<code>format_country(code)</code>	Country name
<code>merge_yf_topics(av_topics, yf_topics)</code>	Combined topic list
<code>get_topic_display_label(key)</code>	Human-readable topic

Consumers: `market_data.py`, `newsroom.py`, `screening.py`, `forecasting.py`, `repository.py`, `components/companyProfile.py`, `excel_export.py`, `cache_manager.py`

excel_export.py — Excel Export

Function	Exports
<code>export_company_profile_excel(...)</code>	Company Profile tab
<code>export_company_profile_simple_excel(...)</code>	Simplified profile
<code>export_stock_quote_excel(...)</code>	Stock quote data
<code>export_company_full_excel(...)</code>	Full company workbook
<code>export_financial_excel(...)</code>	Income/Balance/Cash Flow tabs

Library: openpyxl with Coresight branding

Consumer: `market_data.py`

filing_units.py — Multi-Filing Documents

Supports 8-K, 6-K, DEF14A, and other sub-document types within a filing year.

Function	Role
<code>get_filing_units(ticker, year, doc_type)</code>	List sub-folders from DB or filesystem
<code>get_default_filing_unit(...)</code>	Default unit for display
<code>build_filing_blob_path(...)</code>	Azure blob path builder
<code>get_local_filing_path(...)</code>	Local cache path
<code>has_filing_units(...)</code>	Boolean check

Constant: `FILING_UNIT_DOC_TYPES = {"8-K", "6-K", "DEF14A", "DEFA14A", "DEF 14A", "PRE14A"}`

Consumer: `company_filings.py` (primary)

retailer_forecaster.py — Revenue Forecasting Engine

Class: `RetailerForecaster`

Method	Role
<code>from_dataframe(df)</code>	Construct from revenue history
<code>summary_stats()</code>	Descriptive statistics
<code>backtest()</code>	Holdout-year MAPE evaluation
<code>forecast(periods=5)</code>	Forward projections
<code>run()</code> / <code>run_full_analysis()</code>	Full pipeline

Models (6): linear regression, CAGR, exponential smoothing, Holt, MA trend, weighted average. Best MAPE model selected; ensemble combines top performers.

Consumers: `revenue_forecast_service.py` , `forecast_admin_service.py` , `forecast_refresh_service.py` , forecast scripts

cache_cleaner.py — Bytecode Purge

Function	Role
<code>clean_pycache()</code>	Delete <code>__pycache__</code> under <code>app/</code>
<code>ensure_fresh_code()</code>	Run once per page load
<code>clean_pycache_once()</code>	Idempotent single purge

Sets `PYTHONDONTWRITEBYTECODE=1` at runtime.

Consumer: `earnings_calls.py` (`ensure_fresh_code` at page load)

Note: `main.py` also purges `__pycache__` on every startup.

blob_doc_type_discovery.py — Maintenance Tool

Scans Azure blobs for `doc_type` folder names not in the app's known list. **Not wired into the app** — callable from scripts/REPL.

Function	Role
<code>discover_unknown_doc_types()</code>	Full scan with watermark
<code>get_last_scan_result()</code>	Last scan JSON
<code>format_discovery_report()</code>	Human-readable report

State file: `data/blob_doc_type_scan_state.json`

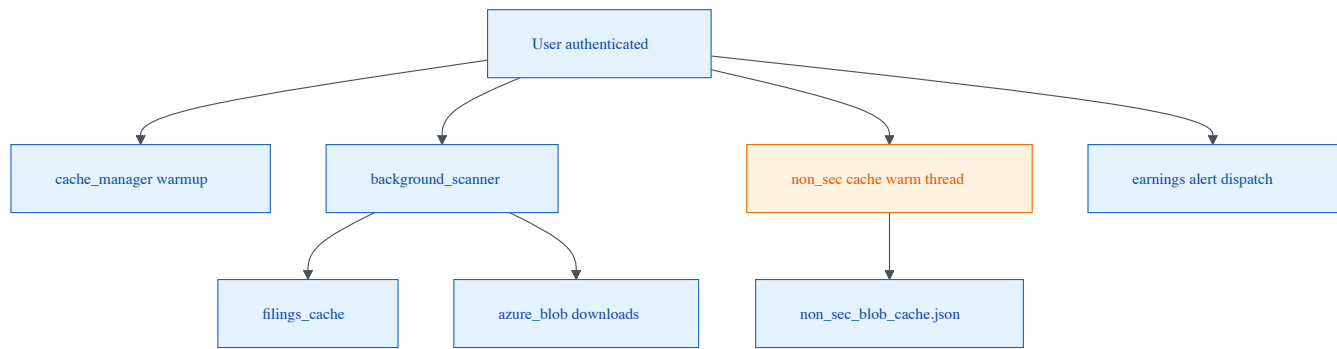
Legacy / Dead Modules

Module	Replacement	Notes
<code>market_data.py</code>	<code>pages/market_data.py</code>	~1,900-line deprecated duplicate (2025-07-10)
<code>filing_pdf.py</code>	<code>html_to_pdf.py</code>	Simple fpdf2 stub
<code>companyProfile.py</code>	<code>components/companyProfile.py</code>	No imports anywhere

Startup Orchestration Summary

Utility	Trigger	Env gate
<code>cache_manager.start_background_warmup</code>	Authenticated session	<code>ENABLE_BG_WARMUP=1</code>
<code>background_scanner.ensure_cache_purge</code> + <code>init_background_scanner</code>	Authenticated session	<code>ENABLE_BG_SCANNER=1</code>
<code>non_sec_blob_fallback.warm_non_sec_cache_at_startup</code>	Authenticated daemon thread	Once per process
<code>earnings_alert_dispatcher</code> tick + thread	Authenticated	<code>EARNINGS_ALERT_DISPATCH=1</code>

Architecture diagram



Dependency Graph (High Level)

```

server_logger ← (used by almost everything)
constants      ← market_data, screening, newsroom, repository, excel_export
azure_blob     ← earnings_calls, non_sec_blob_fallback
filings_cache  ← background_scanner, company_filings
background_scanner → filings_cache, non_sec_blob_fallback
earnings_alert_dispatcher → earnings_alert_email → SMTP
local_storage  ← market_data, toolbar, home
retailer_forecaster ← revenue_forecast_service, forecast_admin_service
  
```